

Tutorial 1

Exercise 1

A consumer has the following utility function:

$$u(x) = (x_1 - b_1)^\alpha (x_2 - b_2)^{1-\alpha}.$$

It represents Stone–Geary preferences. Suppose that $x_1 > b_1 > 0$, $x_2 > b_2 > 0$ and $\alpha \in (0, 1)$.

- (a) Write down the UMP.
- (b) Derive the Marshallian demand function.
- (c) Derive the indirect utility function.
- (d) Check whether demand is homogeneous of degree zero and that it satisfies Walras' law (adding-up).
- (e) Check whether the indirect utility function is homogeneous of degree zero, $\frac{\partial v(p, w)}{\partial w} > 0$ and $\frac{\partial v(p, w)}{\partial p_1} \leq 0$.
- (f) Derive the Marshallian demand function for the following utility function and interpret the result:

$$\tilde{u}(x) = \alpha \ln(x_1 - b_1) + (1 - \alpha) \ln(x_2 - b_2).$$

Exercise 2

- (a) Demonstrate that the budget share of good ℓ , defined by

$$s_\ell(p, w) = \frac{p_\ell x_\ell(p, w)}{w},$$

is increasing after an income increase if good ℓ is a luxury good at (p, w) .

- (b) Derive the income elasticities for the demand derived in Exercise 1 when $w = 1000$, $p_1 = 5$, $p_2 = 1$, $\alpha = 0.3$, $b_1 = 50$, $b_2 = 20$.
- (c) Derive the own-price elasticities for the above demand.
- (d) Derive the cross-price elasticities for the above demand.
- (e) What happens with the above elasticities if $b_1 = b_2 = 0$?

Exercise 3

Consider the utility function

$$u(x) = [\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]^{1/\rho}.$$

This is known as the Constant Elasticity of Substitution (CES) utility function.

- (a) Show that when $\rho = 1$, the indifference curves become linear.
- (b) It can be shown that when $\rho \rightarrow 0$ (and $\alpha_1 + \alpha_2 = 1$), $u(x)$ becomes $x_1^{\alpha_1} x_2^{\alpha_2}$. Which preferences does this function represent?

- (c) When $\rho \rightarrow -\infty$ (and $\alpha_1 = \alpha_2 = 1$), $u(x)$ becomes equal to $\min\{x_1, x_2\}$. Derive the indifference curves.
- (d) Focus on the case $u(x) = [x_1^\rho + x_2^\rho]^{1/\rho}$. Derive the elasticities of substitution for the above three cases. (Derive the formula of the elasticity and fill in respectively $\rho = 1$, $\rho \rightarrow 0$ and $\rho \rightarrow -\infty$ in this formula.)

Exercise 4

Let $x(p, w)$ be a Marshallian demand function defined for all $p \in \mathbb{R}_{++}^L$ and $w > 0$, with $x(p, w) \geq 0$. Suppose that it is differentiable, that it satisfies Walras' law,

$$p \cdot x(p, w) = w \quad \text{for all } (p, w),$$

and that it is homogeneous of degree zero,

$$x(\gamma p, \gamma w) = x(p, w) \quad \text{for all } \gamma > 0.$$

Suppose moreover that there are *no cross-price effects*:

$$\frac{\partial x_\ell(p, w)}{\partial p_k} = 0 \quad \text{for all } k \neq \ell \text{ and all } (p, w).$$

(You may use the following fact without proof: if all partial derivatives of a function $f(p, w)$ with respect to $p_1, \dots, p_L$ are zero at every (p, w) , then f does not depend on p , i.e. $f(p, w) = f(p', w)$ for all p, p' .)

- (a) Prove that the expenditure on each good depends on wealth alone: for every k there is a function g_k such that $p_k x_k(p, w) = g_k(w)$ for all (p, w) .
- (b) Prove that each g_k is linear: $g_k(w) = \bar{s}_k w$ for constants $\bar{s}_k \geq 0$ with $\sum_{k=1}^L \bar{s}_k = 1$.
- (c) Conclude that $x_k(p, w) = \bar{s}_k w / p_k$ for every k , i.e. that demand is of the Cobb–Douglas form with constant budget shares. Exercise 2(e) showed that Cobb–Douglas preferences generate zero cross-price elasticities; state precisely what you have now added.